

Wentao Huang

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EDUCATION

National University of Singapore

Aug 2020 – Feb 2026

- Ph.D. in Computer Science | GPA: 4.70/5.00
- Advisor: Prof. Kian-Lee Tan (ACM Fellow)

Renmin University of China

Sep 2017 – Jun 2020

- M.Phil. in Computer Science | GPA: 3.74/4.00
- Advisor: Prof. Ju Fan

Shandong University

Sep 2013 – Jun 2017

- B.Eng. in Electronic Engineering | GPA: 4.44/5.00 | Rank: 1/39
- Chinese Academy of Sciences (SITP) Elite Class

EXPERIENCE

School of Computing, National University of Singapore

Aug 2024 – Jun 2026

- Student Research Assistant | Principal Investigator: Prof. Kian-Lee Tan

4Paradigm Inc., Singapore

Aug 2023 – Apr 2025

- Ph.D. Student Researcher | Mentors: Dr. Mian Lu, Prof. Bingsheng He (IEEE Fellow)

Illinois Advanced Research Center at Singapore

Jul 2018 – Feb 2019

- Research Engineer | Mentors: Prof. Yuchen Li, Prof. Yuan Fang

PUBLICATIONS

Conference Papers

- [C6] **Wentao Huang**, Qiming Huang, Yunhong Ji, Mian Lu, Yuqiang Chen, Bingsheng He, Kian-Lee Tan. "Taming Multi-Dimensional Skew in Sparse Matrix Multiplication with Contention-Aware Scheduling." 53rd International Conference on Very Large Data Bases (VLDB), 2027. (Accept with Shepherding)
- [C5] **Wentao Huang**, Fanfan Lin. "Quantum-Enhanced Knowledge Graph Systems: Survey, Taxonomy, and the Road to Scalability." 7th International Conference on Quantum Computing and Engineering (QCE), 2026.
- [C4] **Wentao Huang**, Mian Lu, Kian-Lee Tan. "Hash Joins Meet CXL: A Fresh Look." 16th Conference on Innovative Data Systems Research (CIDR), 2026.
- [C3] Yunhong Ji, **Wentao Huang**, Xuan Zhou, Bingsheng He, Kian-Lee Tan. "TaC: An Anti-Caching Key-Value Store on Heterogeneous Memory Architectures." 27th International Conference on Extending Database Technology (EDBT), 2024.
- [C2] **Wentao Huang**, Yunhong Ji, Xuan Zhou, Bingsheng He, Kian-Lee Tan. "A Design Space Exploration and Evaluation for Main-Memory Hash Joins in Storage Class Memory." 49th International Conference on Very Large Data Bases (VLDB), 2023.
- [C1] **Wentao Huang**, Yuchen Li, Yuan Fang, Ju Fan, Hongxia Yang. "BiANE: Bipartite Attributed Network Embedding." 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2020.

Journal Papers

- [J2] Tongfeng Weng, Mo Sha, Xu Zhou, Jingjing Lu, **Wentao Huang**, Kenli Li, Kian-Lee Tan. "SMDG: Enhancing In-Memory Dynamic Graph Processing With Storage-Class Memory." IEEE Transactions on Knowledge and Data Engineering (TKDE), 2026.
- [J1] Yunhong Ji, **Wentao Huang**, Xuan Zhou. "HeterMM: applying in-DRAM index to heterogeneous memory-based key-value stores." Frontiers of Computer Science (FCS), 2024.

Workshop Papers

- [W1] **Wentao Huang**, Mo Sha, Mian Lu, Yuqiang Chen, Bingsheng He, Kian-Lee Tan. "Bandwidth Expansion via CXL: A Pathway to Accelerating In-Memory Analytical Processing." 15th International Workshop on Accelerating Analytics and Data Management Systems Using Modern Processor and Storage (ADMS), 2024.

INVITED TALKS

Scaling Data Infrastructure: Building Bandwidth-Intensive Systems with CXL.	Aug 2026
– Institute of Computing Technology, Chinese Academy of Sciences	
Accelerating Data-Intensive Systems with Emerging Large-Scale Memory Technologies.	Dec 2025
– Shanghai Innovation Institute	
Efficient Analytical Query Processing with Emerging Large-Scale Memory Technologies.	Dec 2025
– Beihang University	

PROFESSIONAL SERVICE

Review Board VLDB'27	2027
Shadow Program Committee Member VLDB'26	2026
Student Area Search Committee NUS School of Computing	2025, 2026
Invited Reviewer FCS'26, ISCAS'24	2024, 2026
Program Committee Member ADMS'25, DL4C@NeurIPS'25	2025
Research Session Deputy Chair EDBT'24	2024
Artifact Evaluation Committee SOSP'23, OSDI'23, USENIX ATC'23, EuroSys'23	2023

PROJECTS

Apache SINGA	Jul 2018 – Aug 2018
– Mentors: Dr. Wei Wang, Prof. Ju Fan Source code: github.com/apache/singa	

TEACHING

CS3223: Database Systems Implementation	Jan 2021 – May 2022
Teaching Assistant, National University of Singapore	
Linear Algebra	Feb 2019 – Jun 2019
Teaching Assistant, Renmin University of China	
Introduction to Computer Science	Feb 2018 – Jun 2018
Teaching Assistant, Renmin University of China	

AWARDS & PRIZES

Research Achievement Award, National University of Singapore	Aug 2023 – Jul 2024
The NUS Research Scholarship, National University of Singapore	Aug 2020 – Jul 2024
The Postgraduate Scholarship, Renmin University of China	Sep 2017 – Jun 2020
The Undergraduate Scholarship, Shandong University	Sep 2013 – Jun 2017
The Challenge Cup, Gold Award, Shandong University	May 2016
Innovation and Entrepreneurship Competition, First Prize, Shandong Province	Oct 2015
The Chinese Mathematics Competition, Third Prize, China	Mar 2015
National Undergraduate Electronic Design Contest, Second Prize, Shandong Province	Sep 2014
University Football Cup, Champion, Shandong University	Apr 2014

MISCELLANEOUS

Open Water Diver, Scuba Schools International	Sep 2018 – Present
Moderator of Technical Q&A, TensorFlow China Community	Sep 2018 – Aug 2019
Minister of English Broadcasting Station, Shandong University	Aug 2014 – Jul 2015